

HONG KONG OBSERVATORY, Hong Kong

WMO: 450050

Lat: 22.302N Lon: 114.174E Elev: 62 StdP: 100.58 Time Zone: 8.00 (E08) Period: 82-92 WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.6	10.9	-1.0	3.5	12.8	1.8	4.3	14.0	9.2	15.9	8.4	15.8	2.2	10	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.5	32.2	26.5	31.7	26.4	31.2	26.3	27.4	30.5	27.1	30.1	26.9	29.9	3.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.6	22.3	29.3	26.2	21.8	29.1	26.1	21.6	29.0	87.5	30.6	86.4	30.3	85.3	30.1	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.6	7.4	6.5		7.6	33.5	1.7	0.3	6.3	33.7	5.3	33.9	4.4	34.1	3.1	34.4	
			WB	4.0	28.2	1.8	0.2	2.7	28.3	1.7	28.4	0.7	28.4	-0.6	28.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.1	16.2	16.1	18.5	21.9	25.8	27.9	29.0	28.8	27.8	25.5	21.7	17.6
	DBStd	5.29	2.37	2.57	3.33	2.70	2.01	1.70	1.29	1.27	1.50	1.85	2.49	2.85
	HDD10.0	2	1	0	1	0	0	0	0	0	0	0	0	1
	HDD18.3	237	72	71	39	3	0	0	0	0	0	0	4	46
	CDD10.0	4782	192	171	264	357	489	538	588	581	534	481	350	236
	CDD18.3	1976	6	8	46	111	231	288	330	323	284	223	104	23
	CDH23.3	18618	0	1	79	436	1826	3226	4110	3924	3126	1657	228	5
	CDH26.7	5853	0	0	0	31	366	1081	1696	1521	957	200	3	0
(14) Wind	WSAvg	3.3	3.2	3.7	3.6	3.5	3.3	3.3	3.1	2.9	3.2	3.6	3.1	2.9
(15) Precipitation	PrecAvg	2225	23	47	69	159	317	383	339	385	300	137	39	31
	PrecMax	3248	101	241	428	492	772	963	1147	872	798	718	145	207
	PrecMin	902	0	0	1	6	6	106	104	85	24	0	0	0
	PrecStd	512	24	62	85	118	202	189	204	220	187	169	43	51
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	23.1	26.0	28.6	31.2	32.2	33.0	32.9	32.5	30.6	26.9	24.0
		MCWB	18.5	19.8	22.7	24.3	26.2	26.5	26.8	26.6	25.6	25.2	21.9	19.3
	2%	DB	20.9	21.9	25.1	27.6	30.2	31.5	32.2	32.1	31.5	29.3	26.0	22.6
		MCWB	17.5	19.3	22.6	24.2	26.0	26.6	26.7	26.6	25.8	24.6	21.7	18.1
	5%	DB	20.0	20.7	24.3	26.7	29.4	30.9	31.6	31.5	30.8	28.5	25.2	21.7
		MCWB	16.9	18.4	22.2	23.8	25.7	26.5	26.6	26.5	25.7	24.2	21.4	17.6
	10%	DB	19.1	19.6	23.3	26.0	28.6	30.3	31.1	31.0	30.1	27.7	24.7	21.0
		MCWB	16.2	17.4	21.4	23.6	25.4	26.4	26.4	26.4	25.5	23.8	21.1	17.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	21.1	23.6	25.3	26.9	27.7	27.7	27.7	27.3	26.2	23.9	20.1
		MCDB	21.0	22.4	25.2	27.1	29.8	30.6	30.9	31.1	30.3	28.9	25.6	22.6
	2%	WB	18.5	19.9	23.1	24.6	26.5	27.2	27.3	27.2	26.9	25.6	23.0	19.4
		MCDB	20.2	21.5	24.7	26.6	29.3	30.0	30.4	30.3	29.6	28.1	25.0	21.8
	5%	WB	17.6	18.9	22.5	24.2	26.1	27.0	27.1	27.0	26.6	25.1	22.4	18.6
		MCDB	19.4	20.5	24.1	26.3	28.8	29.8	30.3	30.0	29.3	27.6	24.6	21.0
	10%	WB	16.8	18.0	21.5	23.8	25.7	26.7	26.9	26.7	26.2	24.6	21.8	17.8
		MCDB	18.7	19.2	23.1	25.9	28.2	29.4	30.0	29.6	29.0	27.2	24.2	20.4
(35) Mean Daily Temperature Range	5% DB	MDBR	3.5	3.1	3.4	3.3	3.3	3.0	3.5	3.5	3.4	3.2	3.5	3.8
		MCDBR	4.1	4.6	4.3	4.2	4.0	3.6	4.1	4.3	4.2	3.7	3.6	4.1
	5% WB	MCWBR	2.7	3.1	2.6	2.2	1.7	1.3	1.3	1.6	1.8	1.9	2.2	2.4
		MCWBR	3.5	4.2	4.0	4.0	3.6	3.3	3.6	3.9	3.7	3.2	3.1	3.6
(40) Clear-Sky Solar Irradiance	taub	taub	0.576	0.627	0.751	0.728	0.567	0.536	0.525	0.590	0.623	0.668	0.570	0.564
		taud	1.814	1.742	1.537	1.622	2.019	2.121	2.149	1.946	1.845	1.697	1.895	1.865
	Ebn at Noon	Ebn at Noon	683	681	620	648	757	774	783	736	702	640	685	675
		Edh at Noon	196	223	285	265	177	159	154	189	206	230	179	180
(44) All-Sky Solar Radiation	RadAvg	RadAvg	3.03	3.11	3.38	3.81	4.54	4.66	5.39	5.03	4.74	4.39	3.53	3.15
	RadStd	RadStd	0.50	0.50	0.45	0.56	0.59	0.69	0.57	0.46	0.37	0.44	0.40	0.46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	+0.25	N/A	N/A	N/A	N/A	N/A	N/A